

# Surfaces of Revolution

## MATH 146 – Calculus II

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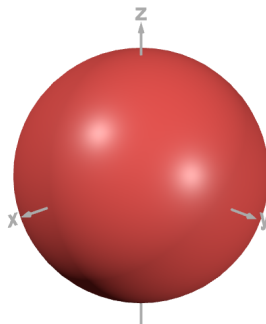
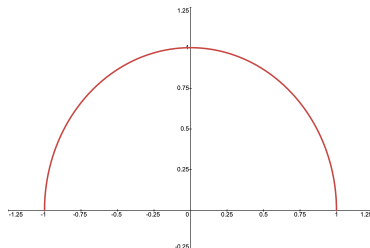


# Surfaces of Revolution

## Surfaces of Revolution

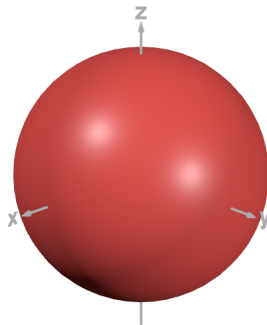
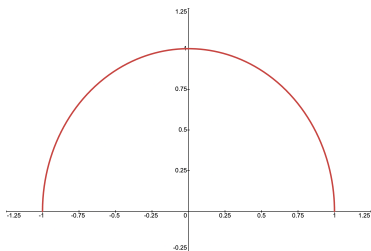
Given a curve  $f(x) \geq 0$  over the interval  $[a, b]$ , a surface of revolution is the 2D figure obtained by revolving the graph of  $f(x)$  around an axis.

$$f(x) = \sqrt{1 - x^2} \text{ over } [-1, 1]$$



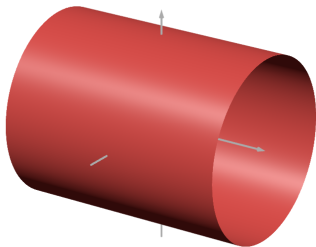
# Common Surfaces of Revolution

$$f(x) = \sqrt{1 - x^2} \text{ over } [-1, 1]$$

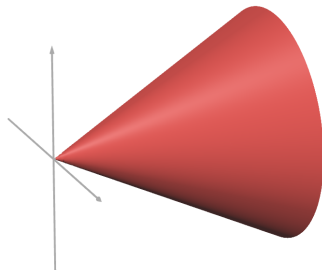


A sphere has a volume of  $V = \frac{4}{3}\pi r^2$  and a surface area of  $SA = 4\pi r^2$ .

# Common Surfaces of Revolution

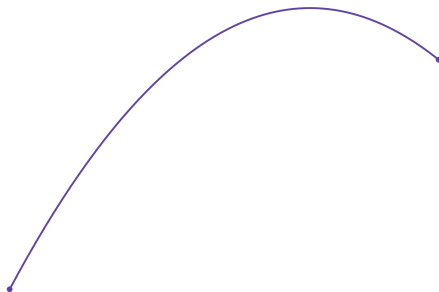


(a)  $SA = 2\pi rh$



(b)  $SA = \pi rl = \pi r\sqrt{r^2 + h^2}$

# Center of Gravity of a Curve



$$\begin{aligned} m_1 &= \sqrt{(x_i - x_{i-1})^2 + (f(x_i) - f(x_{i-1}))^2} \\ &= \sqrt{(x_i - x_{i-1})^2 + f'(x_i^*)^2 (x_i - x_{i-1})^2} \\ &= \sqrt{1 + f'(x_i^*)^2} \Delta x \end{aligned}$$

# Center of Gravity of a Curve

$$\bar{x}_P = \frac{\sum_{i=1}^n m_i \bar{x}_i}{\sum_{i=1}^n m_i} \approx \frac{\sum_{i=1}^n x_i^* \sqrt{1 + f'(x_i^*)^2} \Delta x}{\sum_{i=1}^n \sqrt{1 + f'(x_i^*)^2} \Delta x}$$
$$\bar{y}_P = \frac{\sum_{i=1}^n m_i \bar{y}_i}{\sum_{i=1}^n m_i} \approx \frac{\sum_{i=1}^n f(x_i^*) \sqrt{1 + f'(x_i^*)^2} \Delta x}{\sum_{i=1}^n \sqrt{1 + f'(x_i^*)^2} \Delta x}$$

# Center of Gravity of a Curve

## Theorem

Suppose that  $f(x)$  is continuous and differentiable on the interval  $[a, b]$ . Then the center of gravity of the graph of  $f(x)$  is given by:

$$\bar{x} = \frac{\int_a^b x \sqrt{1 + f'(x)^2} dx}{\int_a^b \sqrt{1 + f'(x)^2} dx} \quad \text{and} \quad \bar{y} = \frac{\int_a^b f(x) \sqrt{1 + f'(x)^2} dx}{\int_a^b \sqrt{1 + f'(x)^2} dx}$$

# Surfaces of Revolution

## Theorem

If a segment of a plane curve  $\mathcal{C}$  is revolved around an axis that does not intersect the curve (except possibly at its endpoints), the area  $SA$  of the resulting surface of revolution is given by the product of the length of  $\mathcal{C}$  times the distance  $d$  traveled by the centroid of  $\mathcal{C}$ .

$$SA = (\text{distance traveled by the center of gravity}) \times (\text{arc length})$$



# Surfaces of Revolution

## Theorem (Revolving a curve around the $x$ -axis)

The surface area of a function  $f(x) > 0$  over  $0 < a < x < b$  when revolved around the  $x$ -axis is:

$$SA = 2\pi \bar{y} \times (\text{arc length})$$

$$\begin{aligned} &= 2\pi \left( \frac{\int_a^b f(x) \sqrt{1 + f'(x)^2} dx}{\int_a^b \sqrt{1 + f'(x)^2} dx} \right) \left( \int_a^b \sqrt{1 + f'(x)^2} dx \right) \\ &= 2\pi \int_a^b f(x) \sqrt{1 + f'(x)^2} dx \end{aligned}$$

# Surfaces of Revolution

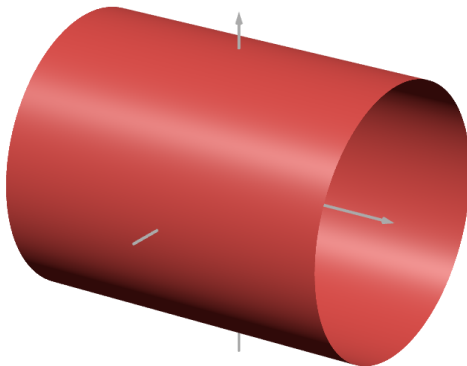
## Theorem (Revolving a curve around the $y$ -axis)

The surface area of a function  $f(x) > 0$  over  $0 < a < x < b$  when revolved around the  $y$ -axis is:

$$\begin{aligned} SA &= 2\pi \bar{x} \times (\text{arc length}) \\ &= 2\pi \left( \frac{\int_a^b x \sqrt{1 + f'(x)^2} dx}{\int_a^b \sqrt{1 + f'(x)^2} dx} \right) \left( \int_a^b \sqrt{1 + f'(x)^2} dx \right) \\ &= 2\pi \int_a^b x \sqrt{1 + f'(x)^2} dx \end{aligned}$$

# Example: Surfaces of Revolution

**Example:** Prove that the (lateral) surface area of a cylinder of height  $h$  and base radius  $r$  is  $SA = \pi r h$ .



# Example: Surfaces of Revolution

**Example:** Prove that the (lateral) surface area of a cone of height  $h$  and base radius  $r$  is  $SA = \pi r l = \pi r \sqrt{r^2 + h^2}$ .

